

TROY MAYER

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EDUCATION

Purdue University

B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics | GPA: 3.4/4.0

West Lafayette, IN

May 2028

EXPERIENCE

Boston Dynamics

Sensors and Perception Co-Op – Atlas

Waltham, MA

July 2026 – Present

- Engineered forward kinematics propagating 6-DOF floating-base poses through multi-joint chains (roll-pitch-yaw matrices, Rodrigues' formula), deriving end-effector velocities by rigid-body velocity composition ($\omega \times r$), not numerical differentiation.
- Fused low-rate state-estimator kinematics with high-rate IMU data into absolute velocity anchors, enabling energy-change estimates across impacts with dead-reckoning drift explicitly bounded.
- Traced under-reported peak impact acceleration to a legacy anti-alias filter; replaced it with a 4th-order zero-phase Bessel filter, chosen over Butterworth after measuring that Butterworth ringing biases windowed impulse integrals.
- Shipped a Flask/Plotly forensics dashboard on a Linux VM over tens of thousands of impact events from robot test logs: ETL from a time-series store into a SQLite cache, per-event waveforms, and a solid-angle-normalized 3D heatmap.
- Designed a Δv severity metric immune to sensor bandwidth limits and measured its noise floor with a null control; gated every refactor on bit-identical regression proofs across the full event table and withheld metrics lacking traceable constants.

Caterpillar (via Purdue Data Mine)

Machine Learning Researcher

West Lafayette, IN

January 2026 – May 2026

- Built Python pipelines merging five years of legacy paint-line files into one daily series and trained an LSTM forecasting next-day use with Monte Carlo intervals; projected 5% less material and contributed to a 4% cycle-time cut.

Chortos Lab – Soft Adaptive Systems and Structures

Undergraduate Researcher

West Lafayette, IN

August 2025 – May 2026

- Wrote a Python computer-vision autotracker resolving cantilever tip displacement to 0.01 mm against an in-frame micrometer scale, and LabVIEW/MATLAB DAQ over 50,000+-point datasets guiding gels with 50%+ more force output.

PROJECTS

EMG-Controlled Active Orthosis – Purdue ASME Biomechanics

Electronics Team Lead (5 engineers)

West Lafayette, IN

April 2024 – Present

- Wrote safety-critical C++17 firmware for an ESP32-S3: a 1 kHz FreeRTOS task running an IMC-tuned PI velocity loop with feed-forward and anti-windup; 10 fault monitors latch the motor off, backed by a hardware watchdog and kill switch.
- Cut muscle-onset-to-motion latency from 103 to 43 ms and stop lag from 89 to 32 ms with a Bayesian EMG estimator, validated in closed-loop simulation on 24 real contractions; 26 GoogleTest tests and an ESP32-S3 build run in CI.
- Redesigned the 4-channel EMG board in KiCad: INA333 front end with right-leg drive, 23–474 Hz band-pass, and ISO7763 isolation plus a hardware kill switch between body-side electronics and the motor driver.
- Emulated both boards on 115 days of real forearm EMG: the redesign held cross-day 11-class accuracy at 87% (v1: 31%); traced a PyTorch CNN's drop to 68% to electrode-polarity flips and recovered 86% with polarity augmentation.

Legged Robot State Estimation – Personal Project

August 2026 – Present

- Implemented a contact-aided error-state EKF in C++/Eigen fusing a 1 kHz IMU with leg kinematics on a simulated quadruped: 0.013 m/s velocity RMSE over a 60-second walk vs. 1.38 m/s for IMU-only dead reckoning, at $\sim 20 \mu s$ per update.
- Verified Jacobians against finite differences in GoogleTest and C++/Python parity to $1e-13$; showed roll, pitch, and velocity stay observable while position and yaw drift, and mapped failure under foot slip and contact latency.

SKILLS

Programming: C++17 (Eigen, GoogleTest, FreeRTOS), Python (NumPy, SciPy, Pandas, PyTorch, scikit-learn), MATLAB, SQL

Robotics & Controls: Rigid-body kinematics, EKF state estimation, PI/feed-forward control, filter design, computer vision

Tools: Linux, Git, GitHub Actions, CMake, Flask, Plotly, SQLite, MuJoCo, KiCad, LabVIEW, NX, Fusion 360, Creo

Languages: English (native), Turkish (fluent), Spanish (fluent, NJ Seal of Bilinguality)